Solving systems

In this lesson, we want to extend much of what we know about solutions of second order equations to solutions of systems of differential equations.

This will actually be a really helpful framework, because much of what we know about the solution of a second order equation is consistent with what we'll see about the solution of a system of equations.

General solution

For instance, remember previously when we learned to solve second order homogeneous equations, that the general solution was given by $y = c_1 y_1 + c_2 y_2$, where $\{y_1, y_2\}$ formed a fundamental set of solutions.

Following this format produced formulas for the general solution with distinct real roots, equal real roots, or complex conjugate roots of the associated characteristic equation.

Roots	General solution
Distinct real roots	$y(x) = c_1 e^{r_1 x} + c_2 e^{r_2 x}$
Equal real roots	$y(x) = c_1 e^{r_1 x} + c_2 x e^{r_1 x}$
Complex conjugate roots	$y(x) = e^{\alpha x}(c_1 \cos(\beta x) + c_2 \sin(\beta x))$ where $r = \alpha \pm \beta i$

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Similarly, the solution to a system of two homogeneous differential equations, written in matrix form as $\vec{x}' = A\vec{x}$, when the matrix A is an 2×2 square matrix, will be given by

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2}$$

This time, instead of the solutions $\{y_1, y_2\}$ which were functions in x, we'll have solutions $\{\vec{x_1}, \vec{x_2}\}$ which are vectors in t. The vectors $\{\vec{x_1}, \vec{x_2}\}$ will be defined as

$$\vec{x_1} = \vec{k_1}e^{\lambda_1 t}$$

$$\vec{x_2} = \vec{k_2}e^{\lambda_2 t}$$

if λ_1, λ_2 are two non-zero real eigenvalues of matrix A. The vectors $\vec{k_1}$ and $\vec{k_2}$ will each have two components, when the matrix A in $\vec{x}' = A\vec{x}$ is a 2×2 square matrix.

And when we're solving a nonhomogeneous system of two equations $\vec{x}' = A\vec{x} + F$, where $F \neq \vec{O}$, the general solution will be the sum of the complementary solution $\vec{x_c} = c_1\vec{x_1} + c_2\vec{x_2}$ (the general solution to the associated homogeneous equation $\vec{x}' = A\vec{x}$ when the matrix A is an 2×2 square matrix) and particular solution $\vec{x_p}$.

$$\vec{x} = \vec{x_c} + \vec{x_p}$$

$$\vec{x} = c_1\vec{x_1} + c_2\vec{x_2} + \vec{x_p}$$

Verifying solutions

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In the past, we learned how to show that a solution set like $\{y_1, y_2\} = \{e^{-3x}, xe^{-3x}\}$ satisfied a second order equation like $y'' + 6y' + 9y = 0$. We simply substituted each solution, $y_1 = e^{-3x}$ and $y_2 = xe^{-3x}$, one at a time into the differential equation, to show that each one individually satisfied the equation.

We can do the same thing with the vector solutions to the system of differential equations. For example, we can show that

$$\overrightarrow{x_1} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} e^{-\frac{3}{2}t}$$

is a solution to the system given in matrix form,

$$\overrightarrow{x}' = \begin{bmatrix} -1 & \frac{1}{4} \\ 1 & -1 \end{bmatrix} \overrightarrow{x}$$

We just need to substitute the solution vector and its derivative into the matrix equation. The derivative of the solution vector is

$$\overrightarrow{x_1}' = -\frac{3}{2} \begin{bmatrix} -1 \\ 2 \end{bmatrix} e^{-\frac{3}{2}t} = \begin{bmatrix} \frac{3}{2} \\ -3 \end{bmatrix} e^{-\frac{3}{2}t}$$

Plugging the solution $\overrightarrow{x_1}$ and its derivative $\overrightarrow{x_1}'$ into the matrix equation gives

$$\begin{bmatrix} \frac{3}{2} \\ -3 \end{bmatrix} e^{-\frac{3}{2}t} = \begin{bmatrix} -1 & \frac{1}{4} \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} e^{-\frac{3}{2}t}$$

$$\begin{bmatrix} \frac{3}{2}e^{-\frac{3}{2}t} \\ -3e^{-\frac{3}{2}t} \end{bmatrix} = \begin{bmatrix} -1 & \frac{1}{4} \\ 1 & -1 \end{bmatrix} \begin{bmatrix} -e^{-\frac{3}{2}t} \\ 2e^{-\frac{3}{2}t} \end{bmatrix}$$

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$$\begin{bmatrix} \frac{3}{2}e^{-\frac{3}{2}t} \\ -3e^{-\frac{3}{2}t} \end{bmatrix} = \begin{bmatrix} (-1)(-e^{-\frac{3}{2}t}) + \left(\frac{1}{4}\right)(2e^{-\frac{3}{2}t}) \\ (1)(-e^{-\frac{3}{2}t}) + (-1)(2e^{-\frac{3}{2}t}) \end{bmatrix}$$

$$\begin{bmatrix} \frac{3}{2}e^{-\frac{3}{2}t} \\ -3e^{-\frac{3}{2}t} \end{bmatrix} = \begin{bmatrix} e^{-\frac{3}{2}t} + \frac{1}{2}e^{-\frac{3}{2}t} \\ -e^{-\frac{3}{2}t} - 2e^{-\frac{3}{2}t} \end{bmatrix}$$

$$\begin{bmatrix} \frac{3}{2}e^{-\frac{3}{2}t} \\ -3e^{-\frac{3}{2}t} \end{bmatrix} = \begin{bmatrix} \frac{3}{2}e^{-\frac{3}{2}t} \\ -3e^{-\frac{3}{2}t} \end{bmatrix}$$

Because we get equivalent values on both sides of the equation, we've verified that

$$\overrightarrow{x_1} = \begin{bmatrix} -1 \\ 2 \end{bmatrix} e^{-\frac{3}{2}t}$$

is, in fact, a solution to the system of differential equations given by the matrix equation.

Superposition, linear independence, and the fundamental set

Furthermore, in the same way that $c_1y_1 + c_2y_2$ is also a solution to the second order equation when $\{y_1, y_2\}$ form a solution set, $c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2}$ is also a solution to the system of differential equations when $\{\overrightarrow{x_1}, \overrightarrow{x_2}\}$ form a solution set. This is the superposition principle.

And when it comes to the solution set $\{\overrightarrow{x_1}, \overrightarrow{x_2}\}$, we're primarily interested in linearly independent solutions, which, just like with solutions to second

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order equations, are solutions that aren't constant multiples of one another.

If we have just two solutions, they'll be linearly independent if they aren't constant multiples of each other. If we have more than two solutions, they'll be linearly independent when none of the solutions can be formed with a linear combination of the other solutions.

Just like before, we can use the Wronskian to test for linear independence.

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \ldots, \overrightarrow{x_n}) = \begin{vmatrix} \vdots & \vdots & \vdots & \vdots \\ \overrightarrow{x_1} & \overrightarrow{x_2} & \cdots & \overrightarrow{x_n} \\ \vdots & \vdots & \vdots & \vdots \end{vmatrix}$$

If the Wronskian of the vector set is non-zero, then the vector set represents a set of linearly independent solutions.

If the Wronskian is non-zero, that means the solution set is linearly independent, and equivalently, we can call the solution set a fundamental set of solutions.

Let's do an example with a homogeneous system to try to tie all these facts together.

Example

Verify that the solution vectors satisfy the system of equations. Calculate the Wronskian of the solution set and use the result to make a statement about the linear (in)dependence of the solutions, and whether or not they form a fundamental set. Then write the general solution of the homogeneous system.

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$$\vec{x}' = \begin{bmatrix} 0 & 6 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \vec{x}$$

$$\{\vec{x_1}, \vec{x_2}, \vec{x_3}\} = \left\{ \begin{bmatrix} 6 \\ -1 \\ -5 \end{bmatrix} e^{-t}, \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} e^{-2t}, \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} e^{3t} \right\}$$

We can verify that each of the vectors $\{\vec{x_1}, \vec{x_2}, \vec{x_3}\}$ are solutions by substituting them into the matrix equation. First though, we'll need the derivative of each solution vector.

$$\vec{x_1} = \begin{bmatrix} 6e^{-t} \\ -e^{-t} \\ -5e^{-t} \end{bmatrix} \qquad \vec{x_1}' = \begin{bmatrix} -6e^{-t} \\ e^{-t} \\ 5e^{-t} \end{bmatrix}$$

$$\vec{x_2} = \begin{bmatrix} -3e^{-2t} \\ e^{-2t} \\ e^{-2t} \end{bmatrix} \qquad \vec{x_2}' = \begin{bmatrix} 6e^{-2t} \\ -2e^{-2t} \\ -2e^{-2t} \end{bmatrix}$$

$$\vec{x_3} = \begin{bmatrix} 2e^{3t} \\ e^{3t} \\ e^{3t} \end{bmatrix} \qquad \vec{x_3}' = \begin{bmatrix} 6e^{3t} \\ 3e^{3t} \\ 3e^{3t} \end{bmatrix}$$

Substitute $\vec{x_1}$ and its derivative $\vec{x_1}'$ into the matrix equation.

$$\begin{bmatrix} -6e^{-t} \\ e^{-t} \\ 5e^{-t} \end{bmatrix} = \begin{bmatrix} 0 & 6 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 6e^{-t} \\ -e^{-t} \\ -5e^{-t} \end{bmatrix}$$

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$$\begin{bmatrix} -6e^{-t} \\ e^{-t} \\ 5e^{-t} \end{bmatrix} = \begin{bmatrix} (0)(6e^{-t}) + (6)(-e^{-t}) + (0)(-5e^{-t}) \\ (1)(6e^{-t}) + (0)(-e^{-t}) + (1)(-5e^{-t}) \\ (1)(6e^{-t}) + (1)(-e^{-t}) + (0)(-5e^{-t}) \end{bmatrix}$$

$$\begin{bmatrix} -6e^{-t} \\ e^{-t} \\ 5e^{-t} \end{bmatrix} = \begin{bmatrix} -6e^{-t} \\ 6e^{-t} - 5e^{-t} \\ 6e^{-t} - e^{-t} \end{bmatrix}$$

$$\begin{bmatrix} -6e^{-t} \\ e^{-t} \\ 5e^{-t} \end{bmatrix} = \begin{bmatrix} -6e^{-t} \\ e^{-t} \\ 5e^{-t} \end{bmatrix}$$

Substitute $\vec{x_2}$ and its derivative $\vec{x_2}'$ into the matrix equation.

$$\begin{bmatrix} 6e^{-2t} \\ -2e^{-2t} \\ -2e^{-2t} \end{bmatrix} = \begin{bmatrix} 0 & 6 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} -3e^{-2t} \\ e^{-2t} \\ e^{-2t} \end{bmatrix}$$

$$\begin{bmatrix} 6e^{-2t} \\ -2e^{-2t} \\ -2e^{-2t} \end{bmatrix} = \begin{bmatrix} (0)(-3e^{-2t}) + (6)(e^{-2t}) + (0)(e^{-2t}) \\ (1)(-3e^{-2t}) + (0)(e^{-2t}) + (1)(e^{-2t}) \\ (1)(-3e^{-2t}) + (1)(e^{-2t}) + (0)(e^{-2t}) \end{bmatrix}$$

$$\begin{bmatrix} 6e^{-2t} \\ -2e^{-2t} \\ -2e^{-2t} \end{bmatrix} = \begin{bmatrix} 6e^{-2t} \\ -3e^{-2t} + e^{-2t} \\ -3e^{-2t} + e^{-2t} \end{bmatrix}$$

$$\begin{bmatrix} 6e^{-2t} \\ -2e^{-2t} \\ -2e^{-2t} \end{bmatrix} = \begin{bmatrix} 6e^{-2t} \\ -2e^{-2t} \\ -2e^{-2t} \end{bmatrix}$$

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Substitute $\overrightarrow{x_3}$ and its derivative $\overrightarrow{x_3}'$ into the matrix equation.

$$\begin{bmatrix} 6e^{3t} \\ 3e^{3t} \\ 3e^{3t} \end{bmatrix} = \begin{bmatrix} 0 & 6 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 2e^{3t} \\ e^{3t} \\ e^{3t} \end{bmatrix}$$

$$\begin{bmatrix} 6e^{3t} \\ 3e^{3t} \\ 3e^{3t} \end{bmatrix} = \begin{bmatrix} (0)(2e^{3t}) + (6)(e^{3t}) + (0)(e^{3t}) \\ (1)(2e^{3t}) + (0)(e^{3t}) + (1)(e^{3t}) \\ (1)(2e^{3t}) + (1)(e^{3t}) + (0)(e^{3t}) \end{bmatrix}$$

$$\begin{bmatrix} 6e^{3t} \\ 3e^{3t} \\ 3e^{3t} \end{bmatrix} = \begin{bmatrix} 6e^{3t} \\ 2e^{3t} + e^{3t} \\ 2e^{3t} + e^{3t} \end{bmatrix}$$

$$\begin{bmatrix} 6e^{3t} \\ 3e^{3t} \\ 3e^{3t} \end{bmatrix} = \begin{bmatrix} 6e^{3t} \\ 3e^{3t} \\ 3e^{3t} \end{bmatrix}$$

Because we found equivalent vectors on both sides of the equation in each of these three cases, we've shown that all three vectors are solutions to the system of differential equations.

Now let's take the Wronskian of the solution set.

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = \begin{vmatrix} 6e^{-t} & -3e^{-2t} & 2e^{3t} \\ -e^{-t} & e^{-2t} & e^{3t} \\ -5e^{-t} & e^{-2t} & e^{3t} \end{vmatrix}$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = 6e^{-t} \begin{vmatrix} e^{-2t} & e^{3t} \\ e^{-2t} & e^{3t} \end{vmatrix} - (-3e^{-2t}) \begin{vmatrix} -e^{-t} & e^{3t} \\ -5e^{-t} & e^{3t} \end{vmatrix} + 2e^{3t} \begin{vmatrix} -e^{-t} & e^{-2t} \\ -5e^{-t} & e^{-2t} \end{vmatrix}$$

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$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = 6e^{-t}[(e^{-2t})(e^{3t}) - (e^{3t})(e^{-2t})]$$

$$+3e^{-2t}[(-e^{-t})(e^{3t}) - (e^{3t})(-5e^{-t})] + 2e^{3t}[(-e^{-t})(e^{-2t}) - (e^{-2t})(-5e^{-t})]$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = 6e^{-t}(e^{t} - e^{t}) + 3e^{-2t}(-e^{2t} + 5e^{2t}) + 2e^{3t}(-e^{-3t} + 5e^{-3t})$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = 6e^{-t}(0) + 3e^{-2t}(4e^{2t}) + 2e^{3t}(4e^{-3t})$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = 12e^{2t-2t} + 8e^{3t-3t}$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = 12(1) + 8(1)$$

$$W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = 20$$

Because the Wronskian is non-zero, $W(\overrightarrow{x_1}, \overrightarrow{x_2}, \overrightarrow{x_3}) = 20 \neq 0$, we can confirm that the vectors in the solution set are linearly independent, which means that the vector set represents a fundamental set of solutions.

Given that this is a fundamental set of solutions, we can write the general solution of the system as

$$\overrightarrow{x} = c_1\overrightarrow{x_1} + c_2\overrightarrow{x_2} + c_3\overrightarrow{x_3}$$

$$\overrightarrow{x} = c_1 \begin{bmatrix} 6 \\ -1 \\ -5 \end{bmatrix} e^{-t} + c_2 \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix} e^{-2t} + c_3 \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix} e^{3t}$$

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